OFFLINE AGENT DEVELOPMENT PROGRESS REPORT

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1 Introduction

The objective of this project is to develop an **OFFLINE AGENT**, an AI-powered assistant capable of interacting with users through both voice and text while performing real-world automation tasks across the system. The focus of the project is to build a locally running intelligent agent that can understand user instructions, maintain conversational context, invoke tools dynamically, and safely execute actions.

The system is currently being developed and tested as a single-agent architecture to validate all core functionalities before transitioning into a multi-agent supervisor-based architecture. Over the past few days, the agent has been upgraded significantly to improve its overall usability, safety, and integration across different system applications.

2 Core Technologies Used

The following programming languages, frameworks, libraries, and tools have been used during the development of the OFFLINE AGENT system.

2.1 Programming Language

- Python as the primary development language for building the agent, automation workflows, and integrations

2.2 AI and LLM Frameworks

- LangGraph for agent workflow orchestration and state-based execution
- LangChain for building tools, tool calling, prompt handling, and agent integration
- Ollama for locally running large language models
- Llama 3.8B as the primary reasoning model currently used

2.3 Memory and Database Components

- SQLite for storing conversational memory and maintaining contextual information
- Context/state handling mechanisms for sequential task execution

2.4 Automation and System Tools

- SMTP libraries for email automation and sending emails
- Python subprocess and shell execution utilities for command execution
- File-system automation utilities for file and directory handling
- Browser and web-search related integrations
- Playwright for browser automation workflows

2.5 User Interaction and Interface Components

- Voice input support for speech-based interaction
- Text-based conversational interface
- Floating window interface for cross-application accessibility
- Yes/No approval mechanism for sensitive operations

3 Implemented Functionalities

3.1 Floating Window System

A floating window interface has been integrated into the project, allowing the agent to remain accessible across multiple applications. This enables seamless interaction with the assistant while working in different environments on the system.

3.2 Tool-Based Action Execution Using LangChain

The agent is capable of performing multiple automation tasks automatically, with all tools built and integrated using LangChain. Some of the currently implemented functionalities include:

- Creating files and directories
- Opening system applications
- Performing system-level automation tasks
- Executing shell commands
- Performing web searches
- Opening YouTube and browser-based applications

3.3 Voice and Text Interaction

The OFFLINE AGENT currently supports both voice-based and text-based interaction. Users can provide instructions naturally, and the agent processes the request before deciding which tools or workflows to invoke.

3.4 Email Automation

Email automation functionality has been implemented successfully using SMTP. The current system can automate email-related actions seamlessly.

3.5 Approval Mechanism for Critical Actions

To improve safety and prevent unintended operations, a Yes/No confirmation step has been added before performing any critical or sensitive action. For tasks such as shell command execution and email sending, the system asks for explicit user confirmation.

3.6 Context Retention and Sequential Tasks

The agent can now retain and use some previous conversational context. It can remember previous interactions and use them during ongoing conversations. While this short-term context handling is functional, I am still working on improving long-term context understanding further.

4 Current Challenges

The following challenges are currently being addressed:

- Improving long-term conversational memory
- Better handling of sequential and multi-step workflows
- Improving reasoning consistency during tool selection
- Optimizing GPU memory usage during local model execution
- Reducing latency and improving execution reliability

During testing, the local LLM setup was observed to consume high GPU memory usage, especially while running Llama 3.8B locally through Ollama. Optimization strategies and lighter model alternatives are currently being evaluated.

5 Planned Enhancements

The following features are planned for future development:

- **Research Feature:** Adding a capability where the agent can research a specified topic, generate a brief summary, and provide the links and sources from which the information was gathered.
- **Context and Sequential Tasks:** Improving the context remembering capability further so the agent can handle long sequential and multi-step tasks more effectively.
- **Voice Command Improvements:** Enhancing voice interaction accuracy and expanding voice-driven capabilities.
- **Advanced Email Features:** Automatic email drafting and intelligent contextual email handling.

6 Future Multi-Agent Architecture

At present, the entire system is implemented as a single agent to simplify development and testing of core functionalities. Once these functionalities become stable, the architecture will be divided into multiple specialized agents working under a supervisor-based architecture.

The planned specialized agents include:

- Web Search Agent
- Coding Agent
- Email Automation Agent
- System Automation Agent

These agents will collaborate under a supervisor agent responsible for task routing, coordination, and workflow management.

7 Conclusion

The OFFLINE AGENT project has successfully achieved several foundational functionalities over the past few days, including a floating window interface, automation tools built via LangChain, SMTP-based email support, approval-based safety mechanisms, multimodal interaction, and basic context retention.

The project is currently progressing toward more advanced capabilities such as improved long-term memory, comprehensive research-oriented workflows, robust multi-step task handling, and multi-agent collaboration.